

Gait-Based Human Identification Using Deep Learning

Group: A Square

Ayan Kumar Batabyal
Ashish Kumar Samantaray

Department of Computer Science

Ramakrishna Mission Vivekananda Educational and Research Institute

30 April 2026

Presentation Flow

- 1 Introduction
- 2 Dataset
- 3 System Pipeline
- 4 Preprocessing
- 5 Model Architecture
- 6 Training
- 7 Evaluation
- 8 Conclusion

Problem Statement

- Human identification using walking style (Gait).
- Non-contact biometric recognition system.
- Useful when face recognition is difficult or obscured.
- Effective in surveillance and security systems.
- Recognition from long distances and low resolutions.

Goal

Identify a person using gait silhouette sequences with high accuracy.

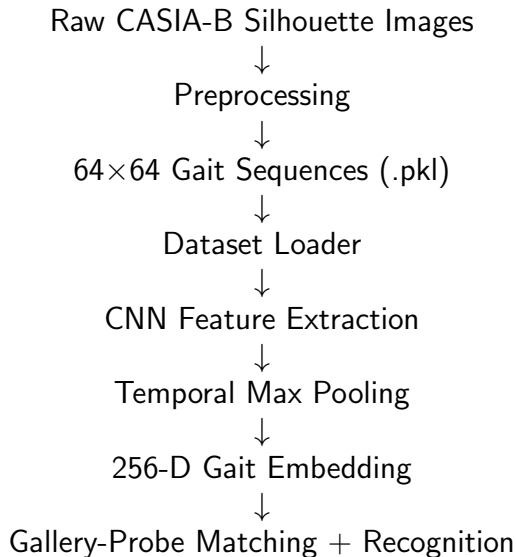
Dataset Overview

- Standard benchmark gait dataset.
- 124 distinct subjects.
- 11 different view angles (000° to 180°).
- 3 distinct walking conditions.

Walking Conditions

- **NM** = Normal Walking
- **BG** = Walking with Bag
- **CL** = Walking with Coat

Complete System Pipeline



Preprocessing Steps

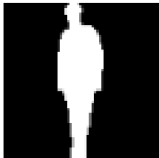
- 1 Read grayscale silhouette image.
- 2 Detect white silhouette pixels.
- 3 Bounding box extraction for precise localization.
- 4 Tight cropping around the subject.
- 5 Aspect ratio preserving resize (prevents distortion).
- 6 Zero padding to a uniform 64×64 tensor.
- 7 Save sequence as a highly optimized .pkl file.

Result

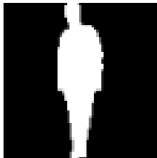
Clean, fixed-size, and biomechanically accurate gait sequences for training.

Preprocessing Visualization

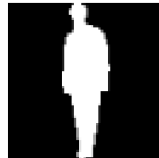
Frame 1



Frame 2



Frame 3



Frame 4



Frame 5



Frame 6



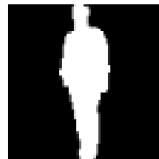
Frame 7



Frame 8

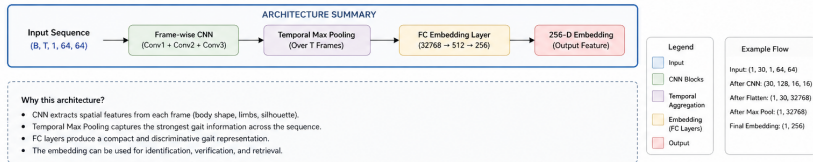
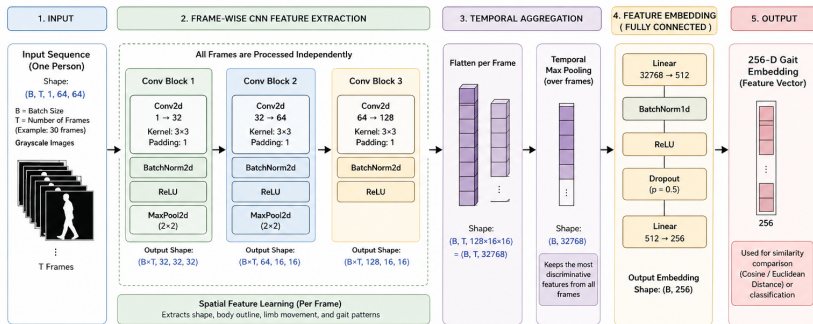


Frame 9



Proposed Model Architecture

Gait Recognition Model Architecture



Model Core Concept

- **Input Shape:** (B, T, 1, 64, 64)
- **Sequence Length:** T = 30 frames per sequence
- **Extraction:** Frame-wise CNN spatial feature extraction
- **Aggregation:** Temporal Max Pooling across frames
- **Compression:** Fully Connected embedding layer

$32768 \rightarrow 512 \rightarrow 256$

Final Output: 256-Dimensional Gait Embedding Vector

$$d_{cos}(x, y) = 1 - \frac{x \cdot y}{\|x\| \|y\|}$$

- **Mathematical Principle:** Smaller distance = Better match.
- **Gallery:** Stores the reference identities (known subjects).
- **Probe:** The unknown walking sequence to be identified.
- **Decision:** The best similarity score dictates the final prediction.

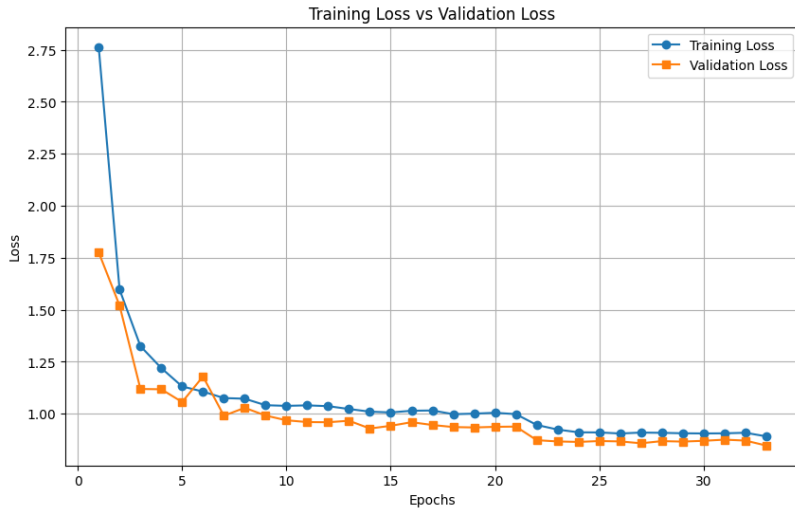
Model Training Configuration

Training Component	Details
Batch Size	16
Learning Rate	0.001
Total Epochs	33
Optimizer	Adam Optimizer
Loss Function	CrossEntropy Loss
Regularization	Dropout + Weight Decay
Learning Rate Scheduler	ReduceLROnPlateau
Early Stopping	Enabled
Best Model Saved At	Epoch 26
Best Validation Accuracy	99.82%

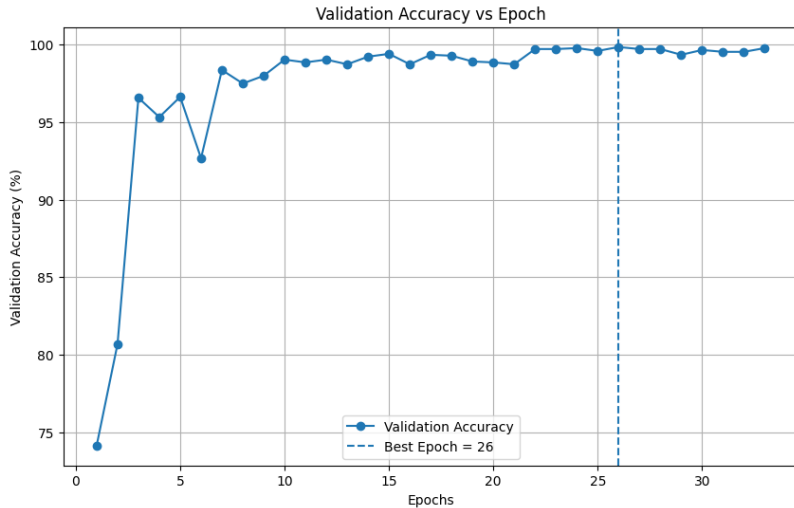
Training Objective

Learn robust 256-dimensional gait embeddings for accurate multi-view person identification and unseen subject recognition.

Training vs Validation Loss



Validation Accuracy Curve



Gallery

- NM-01 @ 090°
- Reference identity database.

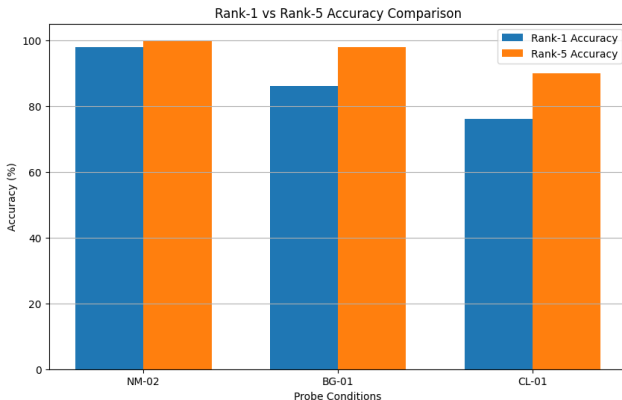
Probe

- NM-02 @ All Angles
- Unknown query sample.

Metrics

- Rank-1 Accuracy
- Rank-5 Accuracy

Condition-wise Performance

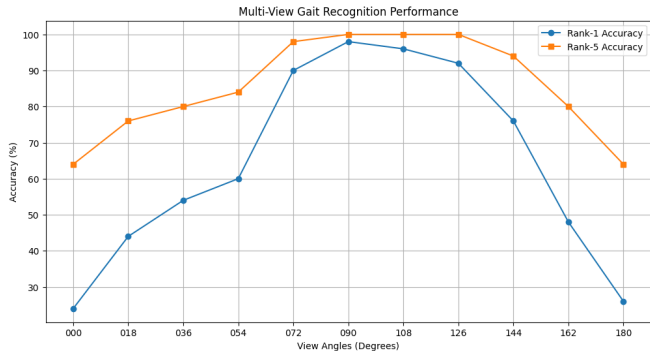


Condition	Rank-1	Rank-5
NM-02	98%	100%
BG-01	86%	98%
CL-01	76%	90%

Observation

The Coat (CL) condition presents the most difficult recognition scenario due to heavy silhouette distortion.

Multi-View Gait Recognition Performance



Angle	000	018	036	054	072	090	108	126	144	162	180
Rank-1	24	44	54	60	90	98	96	92	76	48	26
Rank-5	64	76	80	84	98	100	100	100	94	80	64

Advantages of Proposed System

- Highly robust, aspect-ratio preserving preprocessing pipeline.
- Effective temporal sequence modeling captures stride dynamics.
- Embedding-based recognition provides reliable metric distances.
- Proven capability in Unseen Subject testing.
- Comprehensive multi-view and varying-condition evaluation.

- Implementation of Triplet Loss for superior embedding space separation.
- Integration of Spatial-Temporal Attention mechanisms.
- Exploration of Vision Transformers for gait sequence modeling.
- Benchmarking against GaitSet and GaitPart architectures.
- Real-time CCTV deployment and optimization.
- Advanced Cross-View domain adaptation techniques.

- Successfully developed a complete deep learning-based gait recognition system.
- Leveraged a CNN + Temporal Max Pooling architecture to handle video sequences.
- Generated compact 256-D gait embeddings for fast biometric identification.
- Proved system efficacy via Gallery-Probe matching paradigms.
- Demonstrated strong performance on multi-view, unseen subjects.

Thank You